Predictive analytics

Block 1 - Simple Linear Regression

Ca' Foscari University of Venice

Academic year 2024/2025

This Course

Syllabus

1. Introduction:
   - What is predictive modeling and, more in general, statistical learning?
   - General notation and background
2. Linear models I
   - Model formulation and least squares
   - Inference and prediction
3. Linear models II: model selection, extensions, and diagnostics
4. Generalized linear models (and extension)

- **Warning:** A lot of practicals with **R** - maybe refresh a bit.
- We will use R for data analysis and writing reports using Rmarkdown
- Bring your laptop: we will play with real data!

References

- Textbooks:
  - Julian J. Faraway, 2014.*Linear Models with R Second Edition*, Chapman and Hall/CRC
  - Julian J. Faraway, 2016. *Extending the Linear Model with R: Generalized Linear, Mixed Effects and Nonparametric Regression Models*, Second Edition Chapman and Hall/CRC
  - James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2013. An Introduction to Statistical Learning. Springer Open access book (ISLR book - new edition just out)
  - Peter H. Westfall, Andrea L. Arias, Understanding Regression Analysis - A Conditional Distribution Approach, Chapman and Hall/CRC

- Other teaching materials will be regularly posted on Moodle, i.e.
  - These slides (thanks to Carlo Gaetan for a pevious version of these)
  - Lab sessions
  - A calendar with the list of topics covered during each lesson, so you can keep the pace if you miss a class

Other (free) material

Books which have inspired the course material:

- Lecture notes by Julian Faraway
- The Elements of Statistical Learning by Hastie, Tibshirani and Friedman (the "big brother" of ISLR)
- A draft textbook on data analysis methods by Cosma Shalizi
- Introduction to Data Science by Rafael A. Irizarry
- Applied Statistics with R by David Dalpiaz

There is a deluge of material out there: let me know if you find something good.

The last two books have been written using `bookdown` - an R package which extends `rmarkdown`.

Course Organization

Moodle: https://moodle.unive.it/course/view.php?idnumber=218249

- Lessons:

Day	Time	Room
Monday	10:30 - 12:00	Zeta Aula C
Thursday	10:30 - 12:00	Zeta Aula C

- Office hours: Monday/Thursday 12:10 - 13:30 (book **here**)
    - In person or online - specify your choice when booking.
- Please check the notices on my university webpage/calendar for variations
- The **written exam** using the computer consists of exercises designed to measure
    - the theoretical knowledge of the course topics,
    - the ability to apply them for solving real data problems.

  During the written test the use of books, notes, or electronic media is not allowed (but I provide a R script of reference commands).

Disability and Specific Learning Disorders Service

Students with **disabilities** and/or **SLD** that follow this course are invited to report to the **teacher** and to **Disability and SLD Office** any need in order to optimize the preparation for the exam.

For general information about the services offered by Ca' Foscari:

Disability and SLD Office
`inclusione@unive.it`
041 234 7961

Statistical learning: Introduction and examples[1]

[1] This part is mainly based on ISLR. From this book I have extracted or rephrased sentences.

E-mail example

Goal: construct an automatic spam detector that block spam.
Data: information on 4601 emails, in particular,

- whether was it spam (spam) or not (email);
- the relative frequencies of 57 of the most common words or punctuation marks.

word	george	you	your	hp	free	hpl	!	...
spam	0.00	2.26	1.38	0.02	0.52	0.01	0.51	...
email	1.27	1.27	0.44	0.90	0.07	0.43	0.11	...

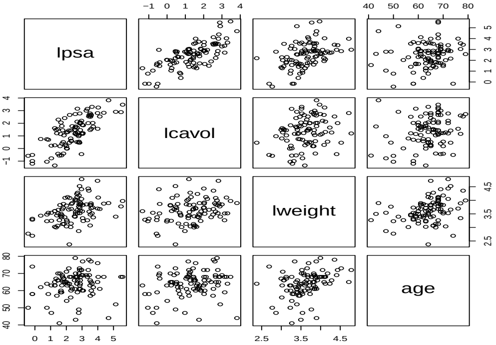
Possible rules:

```
if (%george < 0.6) & (%you > 1.5) then spam else email
```

or

```
if (0.2· %you - 0.3· %george) > 0 then spam else email.
```

Prostate cancer example



- Goal: predict the level of (log) prostate specific antigene (`lpsa`) from some clinical measures, such as log cancer volume (`lcavol`), log prostate weight (`lweight`), age (`age`), ...;
- Possible rule: $\texttt{lpsa} = 0.32 \times \texttt{lcavol} + 0.15 \times \texttt{lweight} + 0.20 \times \texttt{age}$

Statistical Learning

- The goal of statistical learning is to "get knowledge" from the data, so that the information can be used for prediction, identification, understanding, ...
- Statistical learning tools are often divided into
  - supervised
  - unsupervised

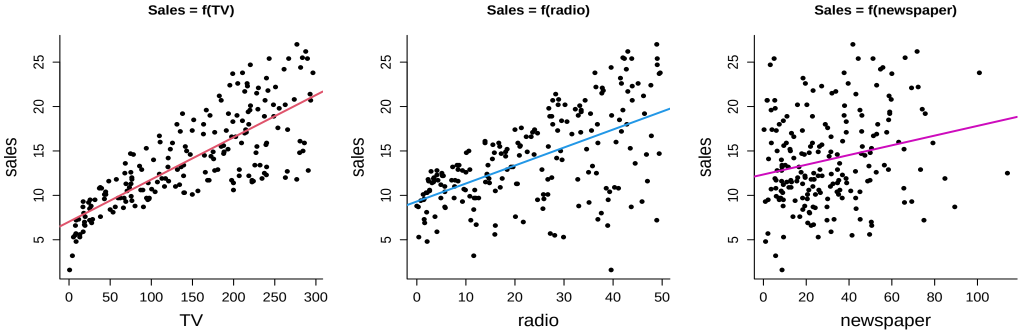
- **Supervised learning**: Given observed data $\{(x_1, y_1), \ldots (x_n, y_n)\}$, learn a statistical model to predict Y from X.
  - If y_i is a continuous numeric value, this task is called prediction (E.g., $Y_i =$ stock price, income, survival time)
  - If y_i is a discrete or symbolic value, this task is called classification (E.g., $y_i \in \{0, 1\}$, $y_i \in \{spam, email\}$, $y_i \in \{1, 2, 3, 4\}$
- **Unsupervised learning**: Given observed data $\{x_1, \ldots, x_n\}$, identify some underlying patterns or structure in the data.

Statistical Learning: examples

- **Supervised learning**
  - Q: To whom should I extend credit?
  - Task: Predict how likely an applicant is to repay loan.
  - Q: How profitable will this subscription customer be?
  - Task: Predict how long customer will remain subscribed.
  - Q: What characterizes customers who are likely to churn?
  - Task: Identify variables that are predictive of churn.
- **Unsupervised learning**
  - Clustering customers into groups with similar spending habits
  - Learning association rules: E.g., 50% of clients, who recently got promoted and had a baby, want to get a mortgage

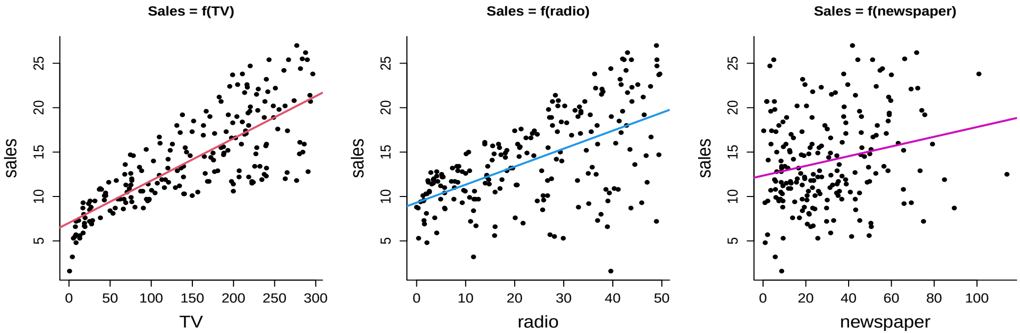
Advertising data set

The Advertising data set contains data on $n = 200$ different markets.



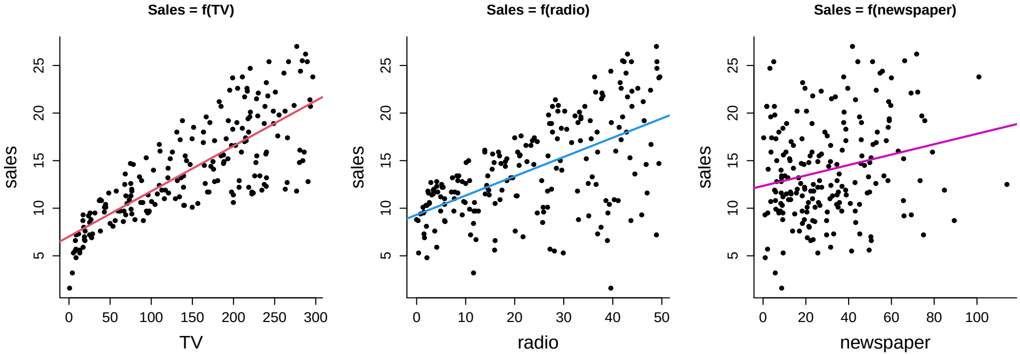
- Outcome $Y = \texttt{sales}$ in 1000's of units
- Inputs: advertising budgets for $X_1 = \texttt{TV}$, $X_2 = \texttt{radio}$, and $X_3 = \texttt{newspaper}$ in 1000's of dollars

Advertising data set



- Ideally, we would like to have a joint model of the form $\texttt{sales} \approx m(\texttt{TV}, \texttt{radio}, \texttt{newspaper})$
- Wanted: a function m such that $m(\texttt{TV}, \texttt{radio}, \texttt{newspaper})$ is a good explanation of $\texttt{sales}$.

Advertising data set



- Here restriced to linear functions.
- What does it mean to be a good predictor?

Notation and Terminology

- `sales` is known as the **response**, or **target**, or **outcome**, or **output**. It's the variable we wish to predict. We denote the response variable as Y.
- `TV` is a **feature**, or **input**, or **predictor**, or **regressor**, or **covariate** or **explanatory variable**.
- We denote `TV` by X_1.
- Similarly, we denote $X_2 =$ `radio` and $X_3 =$ `newspaper`
- Notice that the order of numbering the r.v. is arbitrary.
- We can put all the predictors into a single input vector $\mathbf{X} = (X_1, X_2, X_3)$
- Now we can write our model as

$$Y = m(\mathbf{X}) + \varepsilon$$

  where ε captures **measurement errors** and other discrepancies between the response Y and the model m

Why estimate m ?

- In essence, statistical learning refers to a set of approaches for estimating m starting from the data on X and Y.
- In the sequel, the estimate of m will be denoted by $\widehat{m}(X)$
- Two main reasons why we may wish to estimate m:

  prediction

  inference (interpretation)

Prediction vs inference

- **Prediction**: accurately projecting the chances that something will (or will not) happen.
- Prediction example:

  spam filter: we don't really care why an e-mail filter thinks a message is spam, we only care that the filter works correctly.

- **Inference**: understand why something will (or will not) occur.
- Inference example:

  Which media contribute to sales?

  Which media generate the biggest boost in sales?

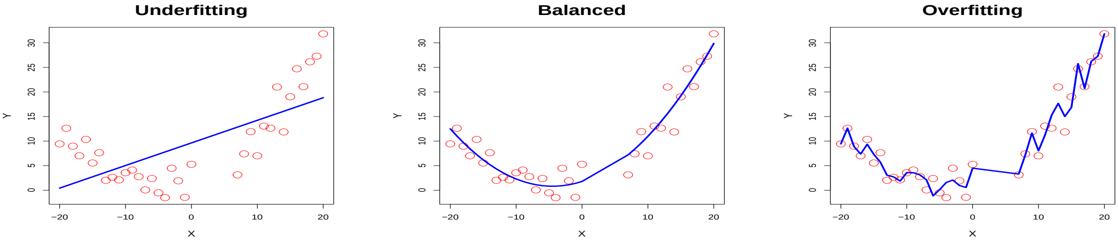
  Size of increase in sales associated to a given increase in TV advertising?

Prediction vs inference

- Sometimes modeling could be conducted both for prediction and inference, it depends on the question.
- Example: in a real estate setting, one may seek to relate values of houses to inputs as crime rate, zoning, distance from a river, etc...
  - Question: is the value of a home given its characteristics under- or over-valued? (prediction problem)
  - Question: what is the additional value for a house to have a view on the river? (inference problem)

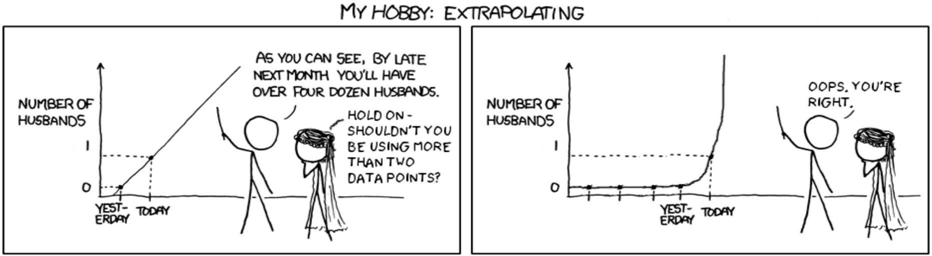
Generalizability

- We want to construct $\widehat{m}$ that generalize well to unseen data
  - Ex. $x = 4$, $\widehat{m}(4)$?
- This is to say, we want predictors that:
  - Capture useful trends in the data (don't underfit)
  - Ignore meaningless random fluctuations in the data (don't overfit)



Generalizability

- We also want to avoid unjustifiably extrapolating beyond the scope of our data

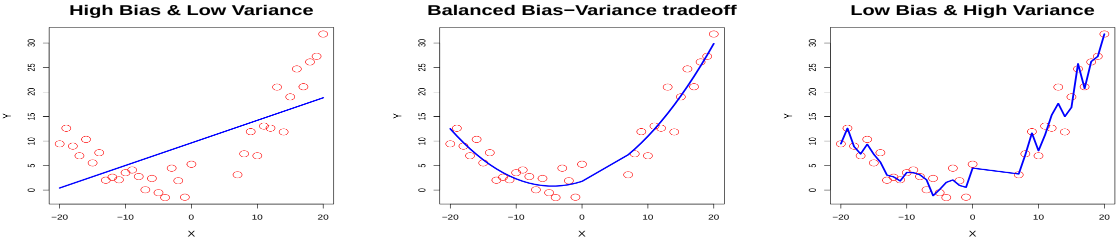


Bias-Variance Tradeoff

- This is encapsulated in the **Bias-Variance tradeoff**, which relates to the fact that given a predictor $\widehat{m}$,

$$\text{Expected-prediction-error}(\widehat{m}) = \text{Variance}(\widehat{m}) + \mathrm{Bias}^2(\widehat{m})$$

- In the language of the previous theme



Interpretability-Flexibility Tradeoff

- We can build highly structured, interpretable models and highly flexible models
- Actually we will focus more on the first ones.
- The best predictor for a problem may turn out to be an uninterpretable or hard-to-interpret black box
- Depending on the purpose of the prediction, we may prefer a more interpretable, worse-performing model to a better-performing "black box".

Interpretability-Flexibility Tradeoff

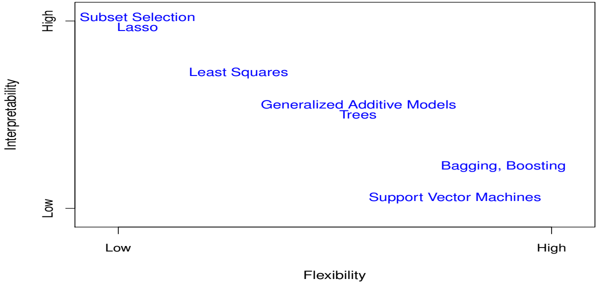


Figure: A representation of the tradeoff between flexibility and interpretability, using different statistical learning methods. In general, as the flexibility of a method increases, its interpretability decreases. Source: ISLR book, fig. 2.7

Reminders from Basic Probability

One random variable

- Expectation of a continuous random variable X with **probability density** function $p(x)$

$$\mathbb{E}[X] = \int xp(x)dx$$

- Expectation of a discrete random variable with probability mass function $p(x)$

$$\mathbb{E}[X] = \sum_x xp(x)$$

- Because everything is parallel for the discrete and continuous cases, I will just write out the integrals

One random variable

- **Expectation** of any function of a random variable $h(X)$

$$\mathbb{E}\left[h(X)\right] = \int h(x)p(x)dx$$

- It is generally the case that $\mathbb{E}\left[h(X)\right] \neq h(\mathbb{E}\left[X\right])$
- $X - \mathbb{E}\left[X\right]$ is the **deviation** or **fluctuation** of X from its expected value.
- The **variance** of X is

$$\mathrm{Var}\left[X\right] = \mathbb{E}\left[(X - \mathbb{E}\left[X\right])^2\right]$$

Bivariate (Multivariate) random variables

- X and Y are two continuous random variables with **joint probability density function** $p_{X,Y}(x,y)$
- the marginal density of X may be recovered by

$$p_X(x) = \int p_{X,Y}(x,y)\,\mathrm{d}y$$

- the **marginal density** of Y may be found with

$$p_Y(y) = \int p_{X,Y}(x,y)\,\mathrm{d}x,$$

Bivariate (Multivariate) random variables

- Two random variables are **independent** if and only if

$$p_{X,Y}(x, y) = p_X(x) \times p_Y(y)$$

- For a function g with arguments (x, y) the expectation of $g(X, Y)$ is

$$\mathbb{E}\left[g(X, Y)\right] = \int g(x, y)\, p_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y,$$

- Set $g(x, y) = (x - \mathbb{E}[X])(Y - \mathbb{E}[Y])$
- The **covariance** of X and Y is

$$\mathrm{Cov}\left[X, Y\right] = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right]$$

Bivariate (Multivariate) random variables

- The covariance is positive when X and Y tend to be above or below their expected values together, and negative if X tend to be above (below) its expected value while Y tends to be below (above) its expected value.
- $\mathrm{Cov}[X, Y] = \mathrm{Cov}[Y, X]$ by definition.
- The **correlation** between X and Y is the covariance between X and Y rescaled to fall in the interval $[-1, 1]$. It is formally defined by

$$\mathrm{Corr}[X, Y] = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}[X]\,\mathrm{Var}[Y]}}.$$

- The correlation will be denoted by $\rho_{X,Y}$ or simply ρ if the random variables are clear from context.
- Important facts about the correlation coefficient:

Bivariate (Multivariate) random variables

- The range of correlation is $-1 \leq \rho_{X,Y} \leq 1$.
- Equality holds above ($\rho_{X,Y} = \pm 1$) if and only if Y is a linear function of X with probability one, i.e. $Y = a + bX$ with probability one.
- If X and Y are independent then $\rho_{X,Y} = 0$, since $\mathrm{Cov}[X, Y] = 0$. The converse is **not** true.
- $\mathrm{Corr}[X, Y] = \mathrm{Corr}[Y, X]$ by definition.

Algebra with expectations, variances and covariances

1. *Linearity of expectations*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

2. *Variance identity*

$$\mathrm{Var}[X] = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2 = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$$

3. *Covariance identity*

$$\mathrm{Cov}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

4. *Covariance is symmetric*

$$\mathrm{Cov}[X, Y] = \mathrm{Cov}[Y, X]$$

Algebra with expectations, variances and covariances

5. *Variance is covariance with itself*

$$\mathrm{Cov}\left[X, X\right] = \mathrm{Var}\left[X\right]$$

6. *Variance is not linear*

$$\mathrm{Var}\left[aX + b\right] = a^2\mathrm{Var}\left[X\right]$$

7. *Covariance is not linear*

$$\mathrm{Cov}\left[aX + b, Y\right] = a\mathrm{Cov}\left[X, Y\right]$$

8. *Variance of a sum*

$$\mathrm{Var}\left[X + Y\right] = \mathrm{Var}\left[X\right] + \mathrm{Var}\left[Y\right] + 2\mathrm{Cov}\left[X, Y\right]$$

9. *Variance of a sum of n variables*

$$\mathrm{Var}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n}\sum_{j=1}^{n}\mathrm{Cov}\left[X_i, X_j\right] = \sum_{i=1}^{n}\mathrm{Var}\left[X_i\right] + 2\sum_{i=1}^{n-1}\sum_{j>i}\mathrm{Cov}\left[X_i, X_j\right]$$

Conditional Distributions

- If x is such that $p_X(x) > 0$, then we define the **conditional density** of $Y|X = x$, denoted $p_{Y|x}$, by

$$p_{Y|x}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)},$$

- We define $p_{X|y}$, the **conditional density** of $X|Y = y$, in a similar fashion, i.e. for y such that $p_Y(y) > 0$

$$p_{X|y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$

Conditional Distributions

- If X and Y are independent

$$p_{X|y}(x|y) = p_X(x), \qquad p_{Y|x}(y|x) = p_Y(y)$$

- The **conditional expectation** of Y given $X = x$ is the expectation computed relative to the conditional distribution i.e.

$$\mathbb{E}\left[Y|X = x\right] = \int y \, p_{Y|x}(y|x) dy$$

- Since $\mathbb{E}\left[Y|X = x\right]$ depends on the given value x of X, we can denote the function

$$\mu(x) = \mathbb{E}\left[Y|X = x\right]$$

- The random variable $\mu(X)$ is called the conditional expected value of Y given X or **regression function** and is denoted by $\mathbb{E}\left[Y|X\right]$

Conditional Distributions

- Similarly we can define the **conditional variance** of Y given $X = x$

$$\begin{aligned}\mathrm{Var}\left[Y|X=x\right] &= \mathbb{E}\left[(Y-\mathbb{E}\left[Y|X=x\right])^2|X=x\right] \\ &= \int (y-\mathbb{E}\left[Y|X=x\right])^2 p_{Y|X}(y|x)dy\end{aligned}$$

- Again, $\mathrm{Var}\left[Y|X=x\right]$ depends on the given value x of X, we can denote the function $\nu(x) = \mathrm{Var}\left[Y|X=x\right]$. The random variable $\nu(X)$ is denoted by $\mathrm{Var}\left[Y|X\right]$

Conditional Distributions

- Useful properties:
  - For a real function r
$$\mathbb{E}\left[r(X)\mathbb{E}\left[Y|X\right]\right] = \mathbb{E}\left[r(X)Y\right]$$
  - **Law of total expectation**. By taking r to be the constant function 1 in the previous formula
$$\mathbb{E}\left[\mathbb{E}\left[Y|X\right]\right] = \mathbb{E}\left[Y\right]$$
  - For any real function r the two random variable $Y - \mathbb{E}\left[Y|X\right]$ and $r(X)$ are uncorrelated i.e.
$$\mathrm{Corr}\left[Y - \mathbb{E}\left[Y|X\right], r(X)\right] = \mathrm{Cov}\left[Y - \mathbb{E}\left[Y|X\right], r(X)\right] = 0$$
  - For any real function r
$$\mathbb{E}\left[r(X)Y|X\right] = r(X)\mathbb{E}\left[Y|X\right]$$
  - **Law of total variance**.
$$\mathrm{Var}\left[Y\right] = \mathrm{Var}\left[\mathbb{E}\left[Y|X\right]\right] + \mathbb{E}\left[\mathrm{Var}\left[Y|X\right]\right]$$

Optimal Prediction [2]

[2] This part is mainly based on a course taught by Cosma Shalizi. From his slides I have extracted or rephrased sentences.

Predict a Random Variable from Its Distribution

- We want to predict the value of a random variable Y.
- What's the best guess we can make?
- We need to measure how good a guess is.
- Let m be our prediction (a scalar).
- The difference $Y - m$ is random and it should somehow be small.
- If we don't care about positive more than negative errors, it's traditional to care about the squared error: $(Y - m)^2$.
- We take its expected value:

$$MSE(m) = \mathbb{E}\left[(Y - m)^2\right]$$

the **Mean Squared Error** of m.

Predict a Random Variable from Its Distribution

- We have the following **bias-variance decomposition**

$$MSE(m) = \mathbb{E}\left[(Y - m)^2\right] = (\mathbb{E}[Y - m])^2 + \mathrm{Var}[Y - m]$$

- $(\mathbb{E}[Y - m])^2$ the squared **bias**
- Remember that $\mathrm{Var}[Y - m] = \mathrm{Var}[Y]$,

$$\begin{aligned} MSE(m) &= (\mathbb{E}[Y] - m)^2 + \mathrm{Var}[Y] \\ &= \text{Bias}^2 + \text{Variance} \end{aligned}$$

Predict a Random Variable from Its Distribution

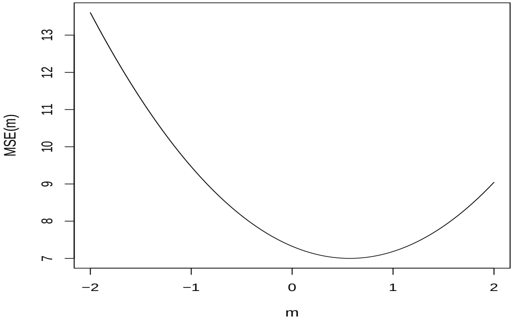


Figure: Mean squared error $\mathbb{E}\left[(Y-m)^2\right]$ as a function of the value m which we predict, when $\mathbb{E}[Y] = 0.57$, $\mathrm{Var}[Y] = 7$.

Predict a Random Variable from Its Distribution

- We wish to pick m such that it minimizes $MSE(m)$ i.e. we want

$$m^* = \mathrm{argmin}_m MSE(m)$$

- Note that the variance term is irrelevant to making this small
- From basic calculus:

$$\frac{dMSE(m)}{dm} = -2(\mathbb{E}[Y] - m) \quad \text{and} \quad \frac{d^2MSE(m)}{dm^2} = 2$$

- First derivative is zero at $m^* = \mathbb{E}[Y]$, a unique minimum.
- So, the best possible one-number prediction m^* for Y is its expected value, i.e.

$$m^* = \mathbb{E}[Y]$$

Predict One Random Variable from Another

- Two random variables, say X and Y.
- We know X and would like to use that knowledge to improve our guess about Y.
- Our guess is therefore a function of x, say $m(x)$.
- We would like

$$\mathbb{E}\left[(Y - m(X))^2\right]$$

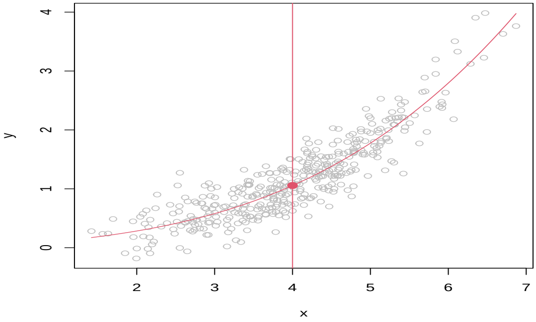
  to be small.
- Use conditional expectations to reduce this problem to the one already solved.

$$\mathbb{E}\left[(Y - m(X))^2\right] \;=\; \mathbb{E}\left[\mathbb{E}\left[(Y - m(X))^2 \mid X\right]\right]$$

- For each possible value x, the optimal value $m^*(x)$ is just the conditional expectation, $\mu(x) = \mathbb{E}\left[Y|X = x\right]$

Predict One Random Variable from Another

The conditional expectation, $\mathbb{E}[Y|X = x]$ is the **regression function** of Y on X.



The Optimal Linear Predictor

- Unfortunately, in general $\mu(x)$ is a really complicated function, for which there exists no nice mathematical expression.
- In searching optimal predictors, we restrict our attention to linear functions $m(x)$ that have the form $b_0 + b_1 x$. (actually, that's an "affine" rather than a "linear" function.)
- The mean squared error of the linear model $b_0 + b_1 x$ is a function of b_0 and b_1 (the parameters) and the variance of Y can be once again ignored. Specifically we find:

The Optimal Linear Predictor

$$MSE(b_0, b_1) = \mathbb{E}\left[(Y - (b_0 + b_1 X))^2\right]$$

The Optimal Linear Predictor

$$
\begin{aligned}
MSE(b_0, b_1) &= \mathbb{E}\left[(Y-(b_0+b_1X))^2\right] \\
&= \mathbb{E}\left[Y^2\right] - 2b_0\mathbb{E}[Y] - 2b_1\mathbb{E}[XY] + \mathbb{E}\left[(b_0+b_1X)^2\right] \\
&= \mathbb{E}\left[Y^2\right] - 2b_0\mathbb{E}[Y] - 2b_1(\mathrm{Cov}[X,Y] + \mathbb{E}[X]\mathbb{E}[Y]) \\
&\quad + b_0^2 + 2b_0b_1\mathbb{E}[X] + b_1^2\mathbb{E}\left[X^2\right] \\
&= \mathbb{E}\left[Y^2\right] - 2b_0\mathbb{E}[Y] - 2b_1\mathrm{Cov}[X,Y] - 2b_1\mathbb{E}[X]\mathbb{E}[Y] \\
&\quad + b_0^2 + 2b_0b_1\mathbb{E}[X] + b_1^2\mathrm{Var}[X] + b_1^2(\mathbb{E}[X])^2
\end{aligned}
$$

The Optimal Linear Predictor

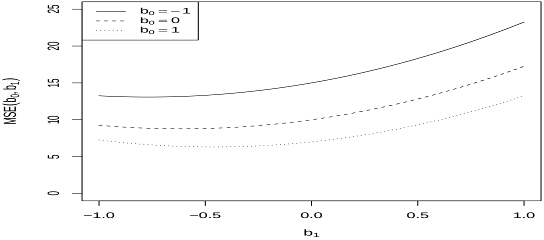


Figure: Mean squared error of linear models with different slopes (b_1) and intercepts (b_0), when $\mathbb{E}[X] = -0.5$, $\mathrm{Var}[X] = 3$, $\mathbb{E}[Y] = 2$, $\mathbb{E}\left[Y^2\right] = 10$, $\mathrm{Cov}[X, Y] = -1$. Each curve represents a different intercept b_0 in the linear model $b_0 + b_1 x$ for Y.

The Optimal Linear Predictor

- We minimize again by setting derivatives to zero

$$\begin{aligned}
\frac{\partial MSE(b_0, b_1)}{\partial b_0} &= -2\mathbb{E}[Y] + 2b_0 + 2b_1\mathbb{E}[X] \\
\frac{\partial MSE(b_0, b_1)}{\partial b_1} &= -2\mathrm{Cov}[X, Y] - 2\mathbb{E}[X]\mathbb{E}[Y] + 2b_0\mathbb{E}[X] \\
&\quad +2b_1\mathrm{Var}[X] + 2b_1(\mathbb{E}[X])^2
\end{aligned}$$

- We denote the optimal value of b_0 and b_1 as β_0 and β_1.

The Optimal Linear Predictor

- From the first equation

$$\beta_0 = \mathbb{E}[Y] - \beta_1 \mathbb{E}[X]$$

Remarks

- The optimal intercept (β_0) enforces the line to go through the mean Y value at the mean X value. (To see this, add $\beta_1 \mathbb{E}[X]$ to both sides.)
- β_0 should have the same units as Y, and the right-hand side of this formula does, because β_1 has the units of Y/X.
- If the variables were "centered", with $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, we'd get $\beta_0 = 0$.

The Optimal Linear Predictor

- Now we plug this in to the other equation:

$$
\begin{aligned}
0 &= -\mathrm{Cov}[X, Y] - \mathbb{E}[X]\mathbb{E}[Y] + \beta_0\mathbb{E}[X] + \beta_1\mathrm{Var}[X] + \beta_1(\mathbb{E}[X])^2 \\
&= -\mathrm{Cov}[X, Y] - \mathbb{E}[X]\mathbb{E}[Y] + (\mathbb{E}[Y] - \beta_1\mathbb{E}[X])\mathbb{E}[X] \\
&\quad +\beta_1\mathrm{Var}[X] + \beta_1(\mathbb{E}[X])^2 \\
&= -\mathrm{Cov}[X, Y] + \beta_1\mathrm{Var}[X]
\end{aligned}
$$

Then

$$\beta_1 = \frac{\mathrm{Cov}[X, Y]}{\mathrm{Var}[X]}$$

The Optimal Linear Predictor

$$\beta_1 = \frac{\mathrm{Cov}\left[X, Y\right]}{\mathrm{Var}\left[X\right]}$$

- Remarks
  - The slope increases the more X and Y tend to fluctuate together, and gets pulled towards zero the more X varies.
  - The expected values $\mathbb{E}\left[X\right]$ and $\mathbb{E}\left[Y\right]$ play no role in the formula, only the variance and covariance matter, and they don't change when we add or subtract constants. In particular, the optimal slope doesn't change if we use $Y - \mathbb{E}\left[Y\right]$ and $X - \mathbb{E}\left[X\right]$ (i.e. the centered variables) in the calculations.
- The line

$$\beta_0 + \beta_1 x$$

is the **optimal regression line** (of Y on X), or the **optimal linear predictor** (of Y on X).

Some important remarks

- We have not assumed that the regression function $\mu(x)$ between X and Y really is linear.
- We have derived the optimal linear approximation to the true relationship, whatever that might be.
- This approximation works in some cases...
- Suppose that $\mu(x)$ is a "sufficiently" smooth function
- Use Taylor series around any value x_0.

$$\mu(x) = \mu(x_0) + (x - x_0)\mu'(x_0) + \frac{1}{2}(x - x_0)^2\mu''(x_0) + \dots$$

Some important remarks

- For x close enough to x_0,

$$\begin{aligned} \mu(x) &\approx \mu(x_0) + (x - x_0)\mu'(x_0) \\ &\approx \mu(x_0) - x_0\mu'(x_0) + \mu'(x_0)x \\ &\approx \beta_0 + \beta_1 x \end{aligned}$$

- How close is enough? Close enough that all the other terms don't matter, so, e.g., the quadratic term has to be negligible, meaning

$$\begin{aligned} |x - x_0|\mu'(x_0) &\gg \frac{1}{2}|x - x_0|^2\mu''(x_0) \quad \text{i.e.} \\ 2\mu'(x_0)/\mu''(x_0) &\gg |x - x_0| \end{aligned}$$

  Unless the function is really straight, therefore, any linear approximation is only going to be good over very short ranges.

Some important remarks

- Think to $\mathbb{E}[Y|X=x] = e^x$, or $\mathbb{E}[Y|X=x] = \sin(x)$.
- Linear models are *computationally* convenient, and there are many situations where computation is at a premium. For instance if you have huge amounts of data, or you need predictions very quickly, or your computing hardware is very weak (say old time computing machines level weak).
- Slogan: *Getting a simple answer can be better than getting the "right" answer*
- No assumptions about the marginal distributions of the variables or about the joint distribution of the two variables
- No assumptions about the fluctuations of Y around the optimal regression line
- No assumption that X came before Y in time, or that X causes Y (what would be the slope of the optimal regression of X on Y?)
- No assumption that X is known precisely but Y with noise, etc.

Some important remarks

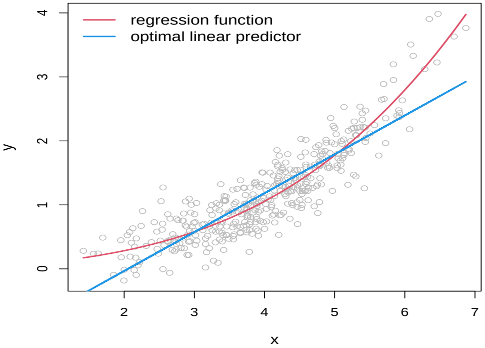


Figure: The regression function $\mathbb{E}[Y|X = x]$ and the optimal linear prediction

Empirical covariance and correlation [3]

[3] This part is mainly based on a book written by Raphael Irizarry. From his book I have extracted or rephrased sentences.

The empirical standard deviation and variance

- Define the empirical mean of a sample $(x_1, \ldots, x_n)$ as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- Define the empirical variance of a sample $(x_1, \ldots, x_n)$ as

$$s^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{1}{n} \left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2 \right)$$

- The empirical standard deviation is defined as $s = \sqrt{s^2}$. Notice that the standard deviation has the same units as the data.

Normalization

- The data defined by
$$z_i = \frac{x_i}{s}$$
have empirical standard deviation 1. This is called "scaling" the data.
- The data defined by
$$z_i = \frac{x_i - \bar{x}}{s}$$
have empirical mean zero and empirical standard deviation 1.
    - The process of centering then scaling the data is called "normalizing" the data.
    - Normalized data are centered at 0 and have units equal to standard deviations of the original data.

The empirical covariance

- Consider now when we have pairs of data, (x_i, y_i).

The empirical covariance

$$c_{XY} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n}\left(\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}\right)$$

Empirical correlation coefficient

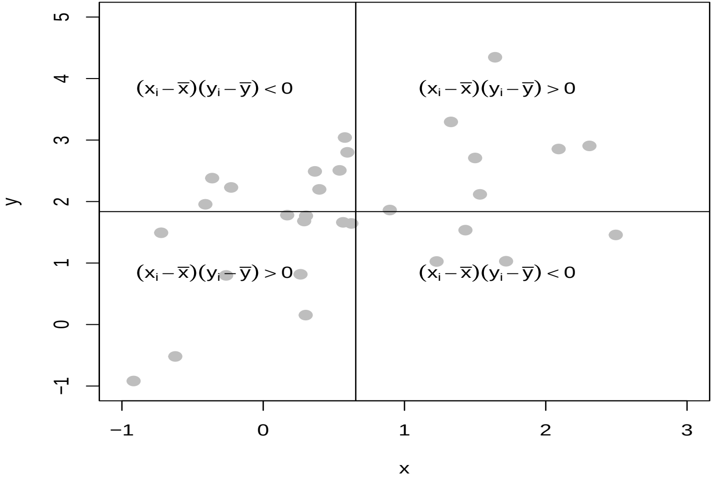
- The correlation coefficient is defined for a sequence of pairs $(x_1, y_1), \ldots, (x_n, y_n)$ as:

$$
\begin{aligned}
r_{XY} &= \frac{1}{n} \sum_{i=1}^{n} \left(\frac{x_i - \overline{x}}{s_X} \right) \left(\frac{y_i - \overline{y}}{s_Y} \right) \\
&= \frac{1}{s_X \, s_Y} \frac{1}{n} \sum_{i=1}^{n} (x_i - \overline{x}) (y_i - \overline{y}) = \frac{c_{XY}}{s_X \, s_Y}
\end{aligned}
$$

  with $\overline{x}, \overline{y}$ the averages of $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ respectively, and s_X, s_Y their standard deviations.

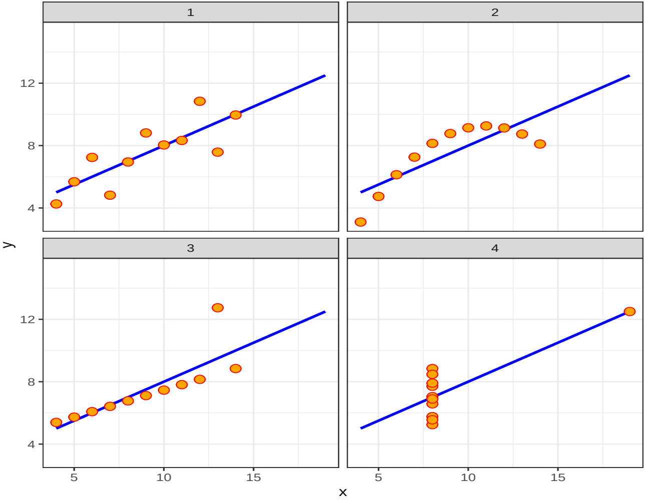
- This correlation coefficients is referred to as the Pearson's correlation coefficient and is based on deviations (i.e. $(x - \overline{x})$ and $(y - \overline{y})$). Other correlation coefficients based on rank also exist (Kendall's τ).

Empirical correlation coefficient



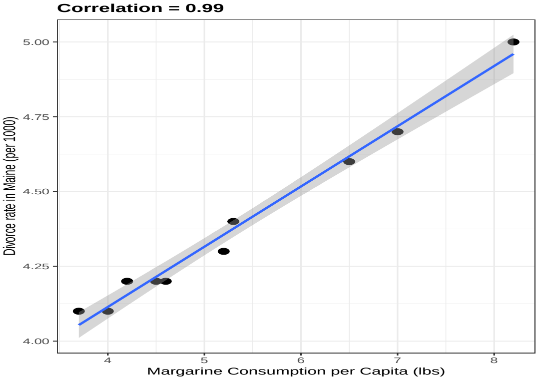
Pearson correlation's pitfalls !

- All these pairs have a correlation of 0.82:



Correlation is not causation

- There are many reasons that a variable X can be correlated with a variable Y without either being a cause for the other.



Correlation is not causation

- Does this mean that margarine causes divorces? Or do divorces cause people to eat more margarine? Of course the answer to both these questions is no.
- This is just an example of what we call a **spurious correlation**.
- You can see many more absurd examples of spurious correlation on this wesbsite

Confounders

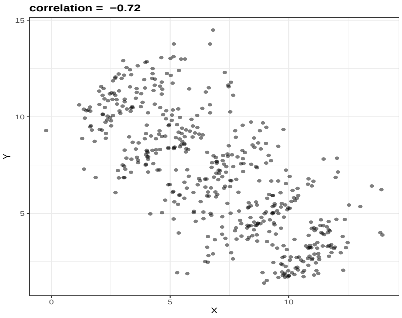
- **Confounders** are perhaps the most common reason that leads to associations being misinterpreted.
- If X and Y are correlated, we call Z a confounder if changes in Z cause changes in both X and Y.
- For example, ice cream sales and murder rates tend to rise together during hot summers. Does that mean that ice cream consumption causes murder, or that murder makes people crave ice cream?

Confounders

- **Confounders** are perhaps the most common reason that leads to associations being misinterpreted.
- If X and Y are correlated, we call Z a confounder if changes in Z cause changes in both X and Y.
- For example, ice cream sales and murder rates tend to rise together during hot summers. Does that mean that ice cream consumption causes murder, or that murder makes people crave ice cream? Of course not.
- Research shows that they both rise in the summer months because warm weather makes ice cream a particularly appealing treat and summer is a time when people are more likely to get together and to be outside, where they come into greater contact with one another.
- Take home meassage: determining causal relationships requires extensive research and subject matter expertise
- Take home meassage 2: observational data are often less than ideal when we wish to make causal statements

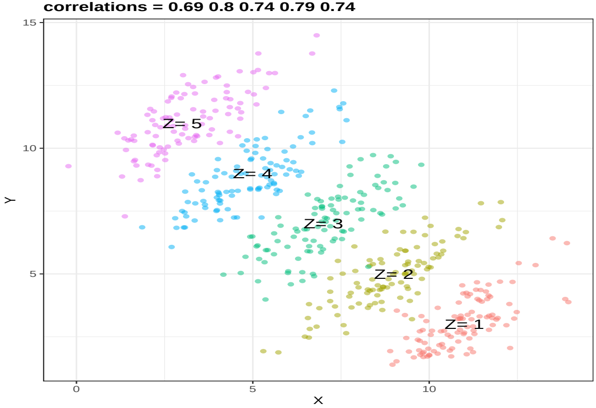
Simpson's Paradox

- It is called a paradox because we see the sign of the correlation of flip when comparing the entire dataset and specific strata.
- Suppose you have three variables X, Y and Z. Here is a scatterplot of Y versus X:



Simpson's Paradox

- X and Y are negatively correlated. However, once we stratify by Z, shown in different colors below, another pattern emerges.



Simpson's Paradox

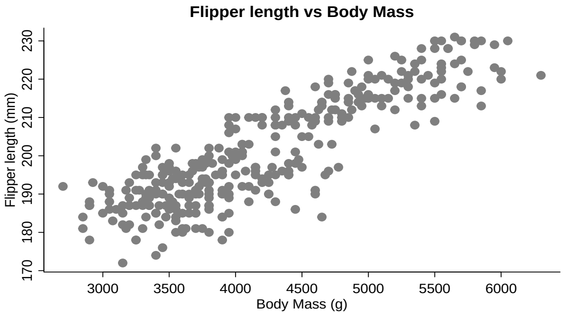
- It is really Z that is negatively correlated with X. If we stratify by Z, the X and Y are actually positively correlated

Simple linear regression model I [4]

[4]Material in based on the Lecture 4 of Cosma Shalizi's course

The Simple Linear Regression Model

- This is a statistical model with two variables X and Y
- We wish to predict/explain Y from X.



- Of all possible functions $Y = m(X)$ we restrict ourselves to a linear (affine) function.

The Simple Linear Regression Model

- In particular we have the following

Assumptions of the model

- The distribution of X is arbitrary (and perhaps X is even non-random).
- $Y = \beta_0 + \beta_1 X + \varepsilon$, for some constants ("coefficients", "parameters") β_0 and β_1, and some random noise variable ε.
- $\mathbb{E}[\varepsilon | X = x] = 0$ (no matter what x is), $\mathrm{Var}[\varepsilon | X = x] = \sigma^2$ (no matter what x is).
- ε is uncorrelated across elements.

- With multiple pairs, $(X_1, Y_1), (X_2, Y_2), \ldots (X_n, Y_n)$, then the model says that, for *each* $i = 1, \ldots, n$,

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where the noise variables ε_i all have the same expectation (0) and the same variance (σ^2), and $\mathrm{Cov}[\varepsilon_i, \varepsilon_j] = 0$ (unless $i = j$, of course).

The Simple Linear Regression Model

- The noise variable may represent measurement error, fluctuations in Y, the effect of the variables which affect Y which we have not (can not) include in the model, etc
- The assumption of **additive** noise is non-trivial — it's not absurd to imagine that either measurement error or fluctuations might change Y multiplicatively (for instance).
- The assumption of a **linear functional form** for the relationship between Y and X is non-trivial; lots of non-linear relationships actually exist.
- The assumption of **constant variance**, or **homoskedasticity**, is non-trivial; the non-correlation assumptions are non-trivial.
- But the assumption that the noise has mean 0 *is* trivial. (Why?)
- Ideally, all of the non-trivial assumptions will be checked, and we will talk later in the course about ways to check them.

Plug-in estimate

- The optimal linear predictor of Y from X has slope and intercept

$$\beta_1 = \frac{\mathrm{Cov}\left[X, Y\right]}{\mathrm{Var}\left[X\right]}, \qquad \beta_0 = \mathbb{E}\left[Y\right] - \beta_1 \mathbb{E}\left[X\right]$$

- $\mathbb{E}\left[Y\right]$, $\mathbb{E}\left[X\right]$ $\mathrm{Cov}\left[X, Y\right]$ and $\mathrm{Var}\left[X\right]$ are functions of the true distribution.
- Rather than having that full distribution, we merely have realizations, say $(x_1, y_1), (x_2, y_2), \ldots (x_n, y_n)$.
- How might we *estimate* β_1 from this empirical data?

Plug-in principle

$$\widehat{\beta_1} = \frac{c_{XY}}{s_X^2}, \qquad \widehat{\beta_0} = \overline{y} - \widehat{\beta_1}\overline{x}$$

Plug-in estimate

- We can't hope that $\widehat{\beta_1} = \beta_1$, but we *can* hope that as $n \to \infty$, $\widehat{\beta_1} \to \beta_1$. i.e. that the estimator is **consistent**,
- Consider (with a slight abuse of notation, note the upper case) the estimators

$$\widehat{\beta_1} = \frac{C_{XY}}{S_X^2}, \qquad \widehat{\beta_0} = \overline{Y} - \widehat{\beta_1}\overline{X}$$

with

$$C_{XY} = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)$$

and

$$S_X^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$$

Plug-in estimate

- Notice that we need $S_x^2 > 0$ for $\widehat{\beta_1}$ to be defined: if there is no variation in X we can not really say much about the effect that X has on Y
- The estimators are consistent if $\overline{Y} \to \mathbb{E}[Y]$, $\overline{X} \to \mathbb{E}[X]$, and $C_{XY} \to \mathrm{Cov}[XY]$, and $S_X^2 \to \mathrm{Var}[X]$.
- On the other hand, it would be nice to say more: we want to know *how far* from the truth our estimate is likely to be, whether it tends to over- or under- estimate the slope, etc.
- We will see in later lectures how the assumptions of the simple linear regression model will let us say something about all of these matters, and how the even stronger assumption that the noise is Gaussian will let us be even more precise.

Least Squares Estimates

- An alternative way of estimating the simple linear regression model
- We keep the assumption that we seek a linear relationship between X and Y: $Y = f(X) = \beta_0 + \beta_1$
- Find β_0 and β_1 which minimize the sum of squared difference between the observed Y_i and the value of Y_i under the model $b_0 + b_i X_i$, i.e.

$$(\beta_0, \beta_1) = \underset{(b_0, b_1)}{\arg\min} \sum_{i=1}^{n} (Y_i - (b_0 + b_1 X_i))^2$$

- The sum of squared deviations corresponds to the

The empirical MSE

$$\widehat{MSE}(b_0, b_1) = \frac{1}{n} \sum_{i=1}^{n} (y_i - (b_0 + b_1 x_i))^2$$

Least Squares Estimates

- If our samples are all independent, for any fixed (b_0, b_1), the law of large numbers tells us that $\widehat{MSE}(b_0, b_1) \to MSE(b_0, b_1)$ as $n \to \infty$.
- So it doesn't seem unreasonable to try minimizing the in-sample error

$$
\begin{aligned}
\frac{\partial \widehat{MSE}}{\partial b_0} &= \frac{1}{n}\sum_{i=1}^{n}(y_i - (b_0 + b_1 x_i))(-2) \\
\frac{\partial \widehat{MSE}}{\partial b_1} &= \frac{1}{n}\sum_{i=1}^{n}(y_i - (b_0 + b_1 x_i))(-2x_i)
\end{aligned}
$$

Set these to zero at the optimum $(\widehat{\beta}_0, \widehat{\beta}_1)$:

Least Squares Estimates

The estimating equations

$$
\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n}\left(y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 x_i)\right) &= 0 \\
\frac{1}{n}\sum_{i=1}^{n}\left(y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 x_i)\right) x_i &= 0
\end{aligned}
$$

i.e.

$$
\begin{aligned}
\overline{y} - \widehat{\beta}_0 - \widehat{\beta}_1 \overline{x} &= 0 \\
\overline{xy} - \widehat{\beta}_0 \overline{x} - \widehat{\beta}_1 \overline{x^2} &= 0
\end{aligned}
$$

Least Squares Estimates

The first equation, re-written, gives

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$$

Substituting this into the remaining equation,

$$\begin{aligned} 0 &= \overline{xy} - \bar{y}\bar{x} + \widehat{\beta}_1 \bar{x}\bar{x} - \widehat{\beta}_1 \overline{x^2} \\ 0 &= c_{XY} - \widehat{\beta}_1 s_X^2 \\ \widehat{\beta}_1 &= \frac{c_{XY}}{s_X^2} \end{aligned}$$

The least square estimates

$$\widehat{\beta_1} = \frac{c_{XY}}{s_X^2}, \qquad \widehat{\beta_0} = \overline{y} - \widehat{\beta_1}\overline{x}$$

Least Squares Estimates

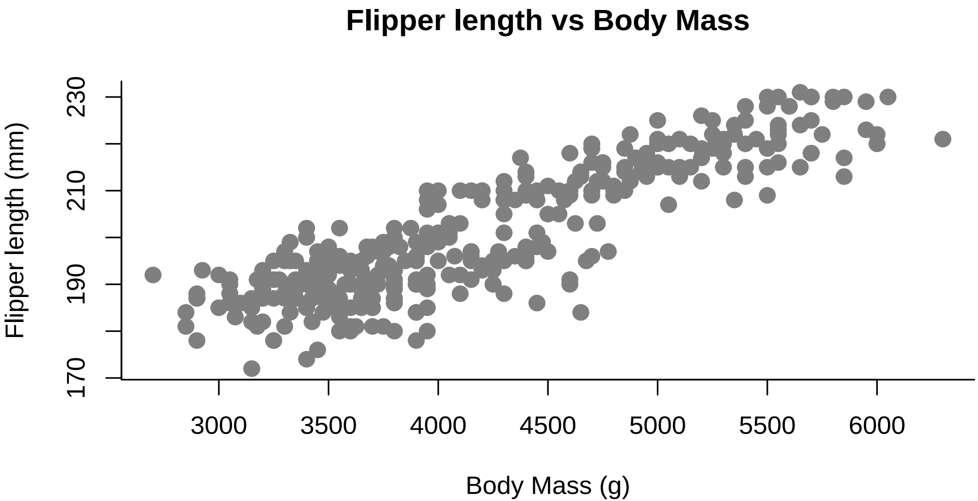
- The equivalence between the plug-in estimates and the least-squares estimates is a bit of a special case for linear models. In some non-linear models, least squares is quite feasible (though the optimum can only be found numerically, not in closed form); in others, plug-in estimates are more useful than optimization.

Penguins

- Palmer's penguins dataset
- How does the size of the penguin (body mass) affects the length of the flipper?
- Let's have a look

```
plot(flipper_length_mm ~ body_mass_g, data = penguins,
  ylab = "Flipper length (mm)",
  xlab = "Body Mass (g)",
  main = "Flipper length vs Body Mass",
  pch  = 20, cex  = 2,col  = "grey50", bty = "l")
```

Penguins


Flipper length vs Body Mass
Flipper length (mm)
170
190
210
230
3000
3500
4000
4500
5000
5500
6000
Body Mass (g)

Function `lm`

- In R we can obtain the least squares estimates using the `lm` function
- Syntax

```
lm(formula = response ~ predictor, data = data)
```

where `response` and `predictor` are the names of two variables/columns in the data frame data (see `?lm`).

```
lm(flipper_length_mm ~ body_mass_g, data = penguins)
```

```
Call:
lm(formula = flipper_length_mm ~ body_mass_g, data = penguins)

Coefficients:
(Intercept)  body_mass_g
   137.0396       0.0152
```

Function `lm`

- The `lm` object

```
fit<-lm(flipper_length_mm ~ body_mass_g, data = penguins);  fit
```

```
Call:
lm(formula = flipper_length_mm ~ body_mass_g, data = penguins)

Coefficients:
(Intercept)  body_mass_g
   137.0396       0.0152
```

- `fit` is a list of objects whose names are

```
typeof(fit)
```

```
[1] "list"
```

Function `lm`

```
names(fit)
```

```
 [1] "coefficients"  "residuals"     "effects"
 [4] "rank"          "fitted.values" "assign"
 [7] "qr"            "df.residual"   "xlevels"
[10] "call"          "terms"         "model"
```

- We can access these elements by $

```
fit$coefficients
```

```
 (Intercept)  body_mass_g
137.03962089   0.01519526
```

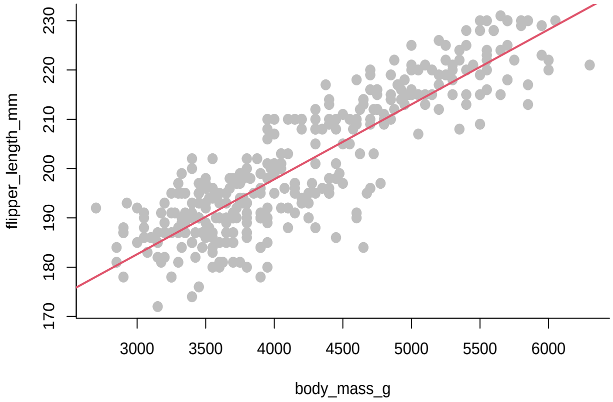
```
fit$call
```

```
lm(formula = flipper_length_mm ~ body_mass_g, data = penguins)
```

Function `lm`

- We can produce a plot with the linear fit easily

Function `lm`



flipper_length_mm
170
180
190
200
210
220
230
3000
3500
4000
4500
5000
5500
6000
body_mass_g

Estimating the slope - a small note

We have seen before that

$$\widehat{\beta_1} = \frac{c_{XY}}{s_X^2}$$

But we know from the definition of the covariance function that: $c_{XY} = r_{XY}\, s_X\, s_Y$ (where r_{XY} is the empirical Pearson correlation coefficient).

We can then write:

$$\widehat{\beta_1} = \frac{r_{XY}\, s_X\, s_Y}{s_X^2} = r_{XY}\frac{s_Y}{s_X}$$

The regression slope is a re-weighting of the Pearson's correlation coefficient which accounts for the variability of the individual variables.

Since s_X and s_Y are positive the sign of $\widehat{\beta_1}$ is the same as the sign of r_{XY}.

Bias and Variance of Parameter Estimates

- It can be shown that the least square estimates are unbiased and have a certain specified variance
- The results are obtained for "designed" or "controlled" experiments, where we get to chose the X values. We therefore derive the conditional expectations $\mathbb{E}\left[\widehat{\beta}_0 | x_1, \ldots x_n\right]$, $\mathbb{E}\left[\widehat{\beta}_1 | x_1, \ldots x_n\right]$ and conditional variances $\mathrm{Var}\left[\widehat{\beta}_0 | x_1, \ldots x_n\right]$, $\mathrm{Var}\left[\widehat{\beta}_1 | x_1, \ldots x_n\right]$. In randomized experiments or in observational studies, obviously the x_i aren't necessarily fixed; however, the results are still correct.
- We have:

$$\mathbb{E}\left[\widehat{\beta}_0 | x_1, \ldots x_n\right] = \beta_0 \quad \text{and} \quad \mathbb{E}\left[\widehat{\beta}_1 | x_1, \ldots x_n\right] = \beta_1$$

Bias and Variance of Parameter Estimates

- first remember that:

$$\sum_{i=1}^{n}(x_i-\overline{x})(y_i-\overline{y}) = \sum(x_iy_i+\overline{xy}-\overline{x}y_i-\overline{y}x_i) =$$

$$\sum x_iy_i + n\overline{x}\ \overline{y} - \overline{x}\sum y_i - \overline{y}\sum x_i = n\overline{x}\ \overline{y} - n\overline{x}\ \overline{y} + \sum(x_iy_i-\overline{x})y_i =$$

$$\sum(x_i-\overline{x})y_i = \sum(y_i-\overline{y})x_i$$

so

$$\widehat{\beta_1} = \frac{c_{xy}}{s_x^2} = \frac{\sum(x_i-\overline{x})(y_i-\overline{y})}{\sum(x_i-\overline{x})^2} = \frac{\sum(x_i-\overline{x})y_i}{\sum(x_i-\overline{x})^2} = \frac{\sum(x_i-\overline{x})(\beta_0+\beta_1x_i+\varepsilon_i)}{(x_i-\overline{x})^2}$$

$$= \frac{\beta_0\sum(x_i-\overline{x})+\beta_1\sum(x_i-\overline{x})x_i+\sum(x_i-\overline{x})\varepsilon_i}{\sum(x_i-\overline{x})^2} = \frac{0+\beta_1\sum(x_i-\overline{x})^2+\sum(x_i-\overline{x})\varepsilon_i}{\sum(x_i-\overline{x})^2}$$

$$= \beta_1 + \frac{\sum(x_i-\overline{x})\varepsilon_i}{\sum(x_i-\overline{x})^2}$$

Bias and Variance of Parameter Estimates

Since $\mathbb{E}\left[\varepsilon_i | X_i\right] = 0$, we have

$$\mathbb{E}\left[\widehat{\beta_1} | x_1, \ldots, x_n\right] = \mathbb{E}\left[\beta_1 + \frac{\sum (x_i - \overline{x})\varepsilon_i}{\sum (x_i - \overline{x})^2} | x_1, \ldots, x_n\right] = \beta_1.$$

It follows that

$$\mathbb{E}\left[\widehat{\beta_0} | x_1, \ldots, x_n\right] = \mathbb{E}\left[\overline{y} - \widehat{\beta_1}\overline{x}\right] = \mathbb{E}\left[\beta_0 + \beta_1 \overline{x} - \widehat{\beta_1}\overline{x} | x_1, \ldots, x_n\right] = \beta_0$$

Bias and Variance of Parameter Estimates

- and we can show that

$$\mathrm{Var}\left[\widehat{\beta}_0 | x_1, \dots, x_n\right] = \sigma^2 \left[\frac{1}{n} + \frac{\overline{x}^2}{n s_X^2}\right] \quad \text{and} \quad \mathrm{Var}\left[\widehat{\beta}_1 | x_1, \dots, x_n\right] = \frac{\sigma^2}{n s_X^2}$$

$$\begin{aligned}
\mathrm{Var}\left[\hat{\beta}_1\right] &= \mathrm{Var}\left[\beta_1 + \frac{\sum(x_i - \overline{x})\varepsilon_i}{\sum(x_i - \overline{x})^2}\right] = \mathrm{Var}\left[\frac{\sum(x_i - \overline{x})\varepsilon_i}{\sum(x_i - \overline{x})^2}\right] = \\
&= \frac{\sum(x_i - \bar{x})^2 \ \mathrm{Var}\left[\varepsilon_i\right]}{\left(\sum(x_i - \overline{x})^2\right)^2} = \frac{n s_X^2 \ \sigma^2}{(n s_X^2)^2} = \frac{\sigma^2}{n s_X^2}
\end{aligned}$$

and

$$\mathrm{Var}\left[\hat{\beta}_0\right] = \mathrm{Var}\left[\overline{y} - \hat{\beta}_1 \overline{x}\right] = \frac{\sigma^2}{n} + \frac{\sigma^2}{n s_X^2}\overline{x}^2$$

Bias and Variance of Parameter Estimates

- Both parameter estimates are unbiased: we can hope that when n is large we recover their real values.
- Looking at the variances: they both have σ^2 at the numerator; and s_x^2 and n at the denominator.
- So, the variability increases when the noise around the regression line increases.
- And the variability decreases as we have more observations (n), which are further spread out along the horizontal axis (s_x^2).

Bias and Variance of Parameter Estimates

- Finally, the **standard error** of an estimator is just its standard deviation, or the square root of its variance:

$$\mathrm{se}(\widehat{\beta}_1) = \frac{\sigma}{\sqrt{ns_X^2}} \quad \text{and} \quad \mathrm{se}(\widehat{\beta}_0) = \sigma\sqrt{\frac{1}{n} + \frac{\overline{x}^2}{ns_X^2}}$$

- It can be shown that the least square estimates are in some sense optimal: they are the minimum-variance mean-unbiased estimators for the simple linear model (when the assumptions are valid).

Unconditional-on-X Properties

- We need to find $\mathbb{E}\left[\widehat{\beta}_1\right]$ and $\mathrm{Var}\left[\widehat{\beta}_1\right]$, not just $\mathbb{E}\left[\widehat{\beta}_1|x_1,\ldots x_n\right]$ and $\mathrm{Var}\left[\widehat{\beta}_1|x_1,\ldots x_n\right]$.
- We use the **law of total expectation**:

$$\begin{aligned}\mathbb{E}\left[\widehat{\beta}_1\right] &= \mathbb{E}\left[\mathbb{E}\left[\widehat{\beta}_1|X_1,\ldots X_n\right]\right]\\ &= \mathbb{E}\left[\beta_1\right] = \beta_1\end{aligned}$$

- That is, the estimator is *unconditionally* unbiased.
- To get the unconditional variance, we use the **law of total variance**:

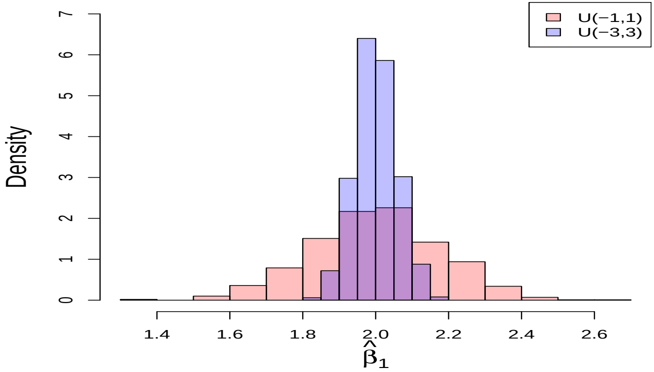
$$\begin{aligned}\mathrm{Var}\left[\widehat{\beta_1}\right] &= \mathbb{E}\left[\mathrm{Var}\left[\widehat{\beta_1}|X_1,\ldots X_n\right]\right] + \mathrm{Var}\left[\mathbb{E}\left[\widehat{\beta}_1|X_1,\ldots X_n\right]\right]\\ &= \mathbb{E}\left[\frac{\sigma^2}{nS_X^2}\right] + \mathrm{Var}\left[\beta_1\right] = \frac{\sigma^2}{n}\mathbb{E}\left[\frac{1}{S_X^2}\right]\end{aligned}$$

Estimator Properties by simulation

- A numerical illustration in which we repeat $m = 1000$ simulation of $n = 100$ observations (x_i, y_i), $i = 1, \ldots, n$ from
$$Y = \beta_0 + \beta_1 X + \varepsilon$$
where $\beta_0 = 1$, $\beta_1 = 2$ and $\varepsilon + 1 \sim \mathrm{Exp}(1)$,i.e. $\mathbb{E}[\varepsilon] = 0$. We consider two setup namely
  - $X \sim U(-1, 1)$
  - $X \sim U(-3, 3)$
- Exercise: Implement this example in R.

Simulation is a powerful way to study the behavior of estimates and models - we will explore this in the R practical sessions.

Estimator Properties by simulation



Estimator Properties by simulation

$\texttt{mean}(\widehat{\beta_0}) = 1$, $\texttt{sd}(\widehat{\beta_0}) = 0.1$; $\texttt{mean}(\widehat{\beta_1}) = 2$, $\texttt{sd}(\widehat{\beta_1}) = 0.18$
$\texttt{mean}(\widehat{\beta_0}) = 1$, $\texttt{sd}(\widehat{\beta_0}) = 0.1$; $\texttt{mean}(\widehat{\beta_1}) = 2$, $\texttt{sd}(\widehat{\beta_1}) = 0.06$

Predictions

- If we knew β_0 and β_1, at $X = x$ our prediction for Y would be

$$\beta_0 + \beta_1 x.$$

- If the simple linear regression model is *true*, i.e. $\mathbb{E}[Y|X = x] = \beta_0 + \beta_1 x$, this is the best prediction we can make.
- However β_0 and β_1 are unknown and we predict that on average Y at an *arbitrary* value of X, say x will be

Point prediction

$$\hat{m}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$$

- But $\hat{m}(x)$ is an *estimate* of $\mathbb{E}[Y|X = x]$, i.e. a function of our data, which are random, hence $\hat{m}(x)$ is also random.

Predictions

- We analyze the randomness in $\hat{m}(x)$.
- First we ask what is its expected value:

$$\mathbb{E}\left[\hat{m}(x)|x, x_1, \ldots x_n\right] = \mathbb{E}\left[\widehat{\beta_0} + \widehat{\beta_1}x|x, x_1, \ldots x_n\right] = \beta_0 + \beta_1 x.$$

- The predictions are unbiased.
- Next we check the variance. To do that we rewrite $\hat{m}(x)$ as:

$$\hat{m} = (\bar{Y} - \widehat{\beta_1}\bar{x}) + \widehat{\beta_1}x = \bar{Y} + (x - \bar{x})\widehat{\beta_1}$$

Predictions

- so we can write:

$$
\begin{aligned}
\operatorname{Var}\left[\hat{m}(x)|x, x_1, \ldots x_n\right] &= \operatorname{Var}\left[\widehat{\beta_0} + \widehat{\beta_1} x | x, x_1, \ldots x_n\right] \\
&= \operatorname{Var}\left[\bar{Y} + (x - \bar{x})\widehat{\beta_1} | x, x_1, \ldots x_n\right] \\
&= \frac{\sigma^2}{n} + (x - \bar{x})^2 \frac{\sigma^2}{n s_x^2} \\
&= \frac{\sigma^2}{n}\left(1 + \frac{(x - \overline{x})^2}{s_X^2}\right).
\end{aligned}
$$

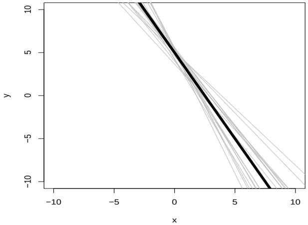
Predictions

$$\mathrm{Var}\left[\hat{m}(x)|x, x_1, \ldots x_n\right] = \frac{\sigma^2}{n}\left(1 + \frac{(x - \overline{x})^2}{s_X^2}\right).$$

- The variance grows as σ^2 grows: the noisier the data the more variable the estimated regression line, and the more of that noise will propagate into our predictions.
- The larger n is, the smaller the variance: the more points we see, the more exactly we can pin down the line, and so our predictions.
- The variance of our predictions is the sum of two terms. The first term is σ^2/n: the variance of $\overline{Y}$. Since our line has to go through the center of the data, this just how much noise there is in the height of that center.
- The other term does change with x, specifically with $(x - \overline{x})^2$: the further our operating point x is from the center of the data $\overline{x}$, the bigger our uncertainty.

Predictions

- The Figure illustrates the spread in point predictions as we move away from $\overline{x}$. We plot several estimated least-squares regression lines (grey) and the true regression line (black).



- Notice how the estimated lines become more spread out as we move away from the mean of the distribution of X (here set at 0)
- Again, the previous equation are conditional on the x_i. If those are random, we need to use the law of total variance to get the unconditional variance of $\hat{m}(X)$.

Estimating σ^2: Sum of Squared Errors

- Under the simple linear regression model, it is easy to show that

$$\mathbb{E}\left[(Y - (\beta_0 + \beta_1 X))^2\right] = \sigma^2 \tag{1}$$

- This suggests that the minimal value of the in-sample MSE,

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{m}(x_i))^2 \tag{2}$$

  is a natural estimator for σ^2.
- This is, in fact, a **consistent** estimator. (You can prove this using the consistency of $\hat{\beta}_0$ and $\hat{\beta}_1$, and continuity.)

Estimating σ^2: Sum of Squared Errors

- It is, a slightly **biased** estimator. Specifically

$$s_e^2 = \frac{n}{n-2}\hat{\sigma}^2.$$

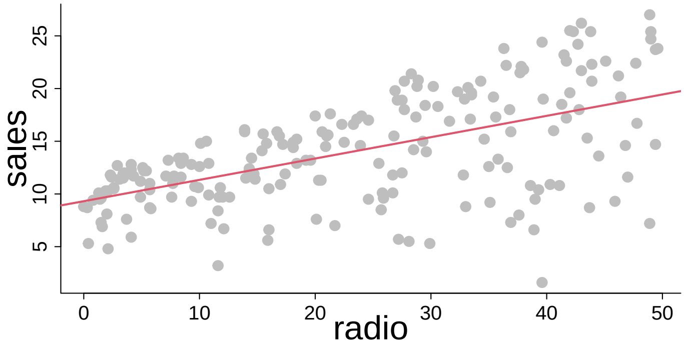
It turns out that S_e^2 is an *un*-biased estimator of σ^2, though one with a larger variance.

Sum of squared errors

$$SSE = \sum_{i=1}^{n}(y_i - \hat{m}(x_i))^2 = \sum_{i=1}^{n}(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

Advertising data

- Let's use the `Advertising` data set and study the relationship between `radio` and `Sales`.



Residuals

The residuals

$$e_i = y_i - \hat{m}(x_i) = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \tag{3}$$

- This may look like re-arranging the basic equation for the linear regression model,

The errors

$$\varepsilon_i = Y_i - (\beta_0 + \beta_1 X_i) \tag{4}$$

  and it *is* similar, but it's *not* the same.

- The right-hand side of Eq. 4 involves the true parameters.
- The right-hand side of Eq. 3 involves the *estimated* parameters, which are different.
- Therefore: The residuals are not the noise terms; $e_i \neq \varepsilon_i$, but ...

Residuals

- ... some ways in which the residuals are *like* the noise terms.
  - The residuals are always uncorrelated with the x_i:

$$\frac{1}{n}\sum_{i=1}^{n} e_i(x_i - \overline{x}) = 0 \tag{5}$$

However, this is a consequence of the estimating equations: it is true *whether or not* the simple linear regression model is actually true.

$$\frac{1}{n}\sum_{i=1}^{n} e_i = 0 \tag{6}$$

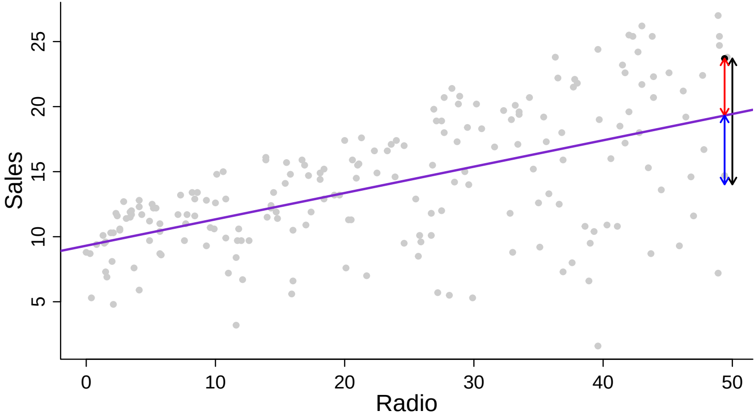
Also a consequence of the estimating equations. This is *reminiscent* of $\mathbb{E}[\varepsilon] = 0$, but generally $n^{-1}\sum_{i=1}^{n}\varepsilon_i \neq 0$.

Residuals

- Despite these differences, there is enough of a relationship between the ε_i and the e_i that a lot of our model-checking and diagnostics will be done in terms of the residuals.

The sum of squares

- How to judge the goodness of fit in a synthetic way?



- Decomposition of the sum of squares; $\mathrm{SS}_{\mathrm{tot}} = \mathrm{SS}_{\mathrm{reg}} + \mathrm{SS}_{\mathrm{res}}$

The sum of squares

- Ingredients (define $\hat{y}_i = \hat{m}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$):
  - the **total sum of squares**

$$\mathrm{SS}_{\mathrm{tot}} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

    this is n times the variance of y, $s_Y^2 = \mathrm{SS}_{\mathrm{tot}}/n$
  - the **explained sum of squares**

$$\mathrm{SS}_{\mathrm{reg}} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

  - the **residual sum of squares**

$$\mathrm{SS}_{\mathrm{res}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} e_i^2 = (n-2)s_e^2$$

The sum of squares

Derived from:

$$
\begin{aligned}
SS_{tot} &= \sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(y_i - \bar{y} + \hat{y}_i - \hat{y}_i)^2 \\
&= \sum_{i=1}^{n}\left\{(y_i - \hat{y}_i)^2 + (\hat{y}_i - \bar{y})^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - \bar{y})\right\} = \\
&= SS_{Res} + SS_{Reg} + 2\sum_{i=1}^{n}(y_i - \hat{y}_i)(\widehat{\beta_0} + \widehat{\beta_1}x_i - \bar{y}) \\
&= SS_{Res} + SS_{Reg} + 2(\widehat{\beta_0} - \bar{y})\sum_{i=1}^{n}(y_i - \hat{y}_i) + 2\widehat{\beta_1}\sum_{i=1}^{n}(y_i - \hat{y}_i)x_i \\
&= SS_{Res} + SS_{Reg}
\end{aligned}
$$

where the last two terms cancel out because of the estimating equations.

The coefficient of determination

- The **coefficient of determination** R^2 is the proportion of variability in the response that is accounted for by the statistical model

R^2 index

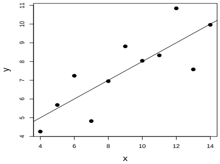
$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = \frac{SS_{\text{reg}}}{SS_{\text{tot}}}$$

- Properties of R^2:
  - R^2 assumes values between 0 and 1
  - the better the model, the smaller the residual sum of squares
- In our example, the R^2 is 0.332
- The latter value is quite high but is not enough for judging the quality of the goodness of fit.

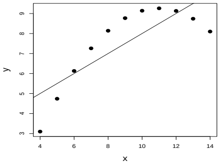
Exercise: show that for the simple linear regression R^2 is exactly the squared value of the correlation coefficient (r)

Inspecting the residuals

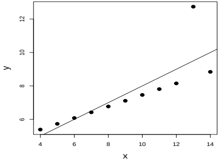
The same "quality" of fitting ...



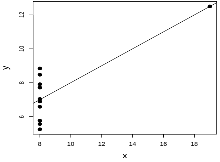
$R^2 = 0.667$



$R^2 = 0.666$



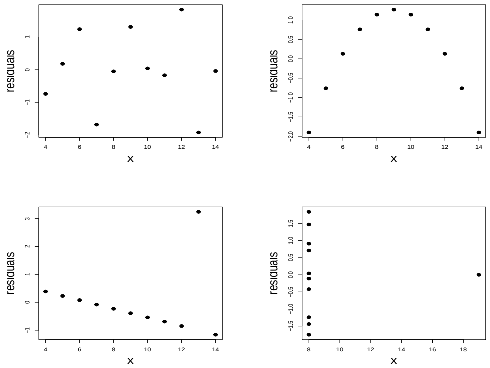
$R^2 = 0.666$



$R^2 = 0.667$

Inspecting the residuals

... but different residuals



Summary on properties of the residuals

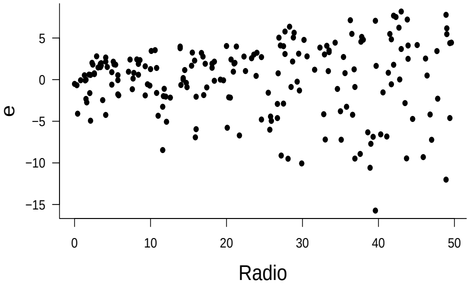
Let's sum up the most relevant observations from the last slides:

1. The residuals should have expectation zero, conditional on x, $\mathbb{E}\left[e_i|X=x\right]=0$. (The residuals should also have an over-all sample mean of exactly zero.)
2. The residuals should show a constant variance, unchanging with x.
3. The residuals can't be completely uncorrelated with each other, but the correlation should be extremely weak, and grow negligible as $n \to \infty$.

Each one of these points leads to a diagnostic, and something to check for in a model.

Plot of the residuals against the predictor X

- This allows us to zoom in on instances of poor model fit. Whenever we look at a residual plot, we are searching for systematic patterns of any sort.
- Here's the plot for Advertisement data:



Plot of the residuals against the predictor X

- Some issues with the constant variance assumption
- The covariance between e_i and x:

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^{n}\left(y_i-(\widehat{\beta}_0+\widehat{\beta}_1x_i)\right)(x_i-\overline{x}) \quad &= \\ \frac{1}{n}\sum_{i=1}^{n}\left(y_i-(\widehat{\beta}_0+\widehat{\beta}_1x_i)\right)x_i \quad &- \quad \frac{1}{n}\sum_{i=1}^{n}\left(y_i-(\widehat{\beta}_0+\widehat{\beta}_1x_i)\right)\bar{x}\end{aligned}$$

- Because of the estimating equation, these are 0 by definition, so

$$\frac{1}{n}\sum_{i=1}^{n}\left(y_i-(\widehat{\beta}_0+\widehat{\beta}_1x_i)\right)(x_i-\overline{x})=\frac{1}{n}\sum_{i=1}^{n}e_i\,(x_i-\overline{x})=0$$

i.e. the residuals are uncorrelated with the predictor, but it doesn't mean that the residuals can not exhibit non linear dependence with the predictor. If non-linearities are visible - the relationship between X and Y might be more complex than assumed.

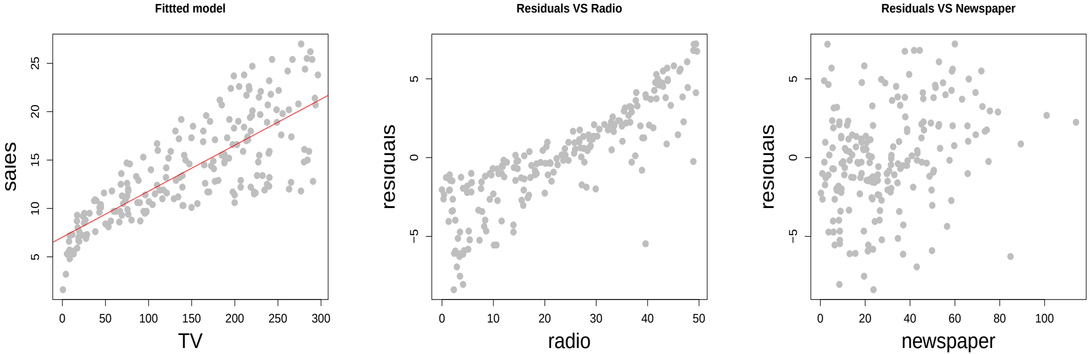
Plotting against another variable...

- If you have other potential predictor variables, you should be able to plot the residual against them and see a flat line around zero. If not, that is an indication that the other variable might indeed help predict the response, so it should probably be incorporated in your model.
- In particular, if you make such a plot and you see the points in it fall around a straight line, that's an strong indication that you need a **multiple linear regression model**.
- Example: advertising data. We want to predict `sales` using `TV`

$$\texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \varepsilon$$

- ... but other variables (`radio`, `newspaper`) could be potential predictors

Plotting against another variable...

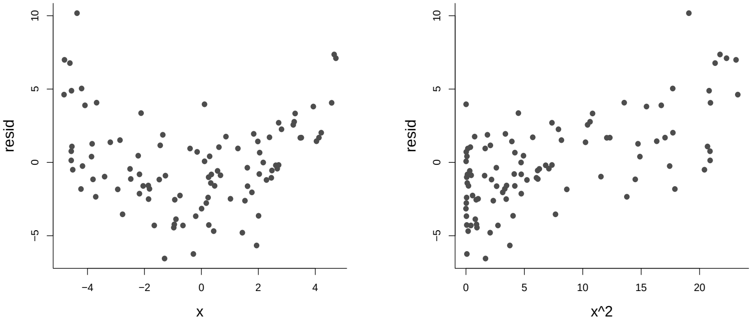


- In this case `radio` could be considered, i.e.

$$\texttt{sales} = \beta_0 + \beta_1 \texttt{TV} + \color{blue}{\beta_2 \texttt{radio}} + \varepsilon$$

Plotting against another variable...

- The other variable might be a transformation of the original variable (fictional data)



Plotting against another variable...

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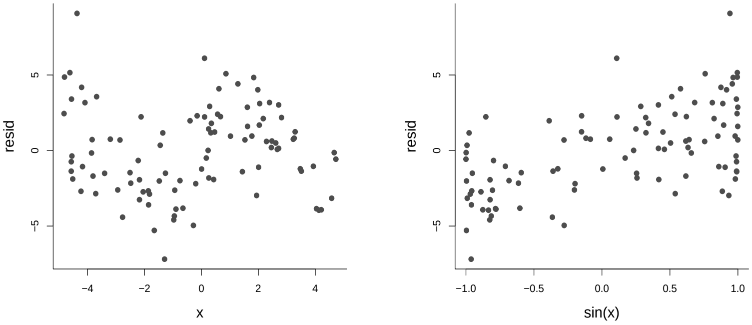


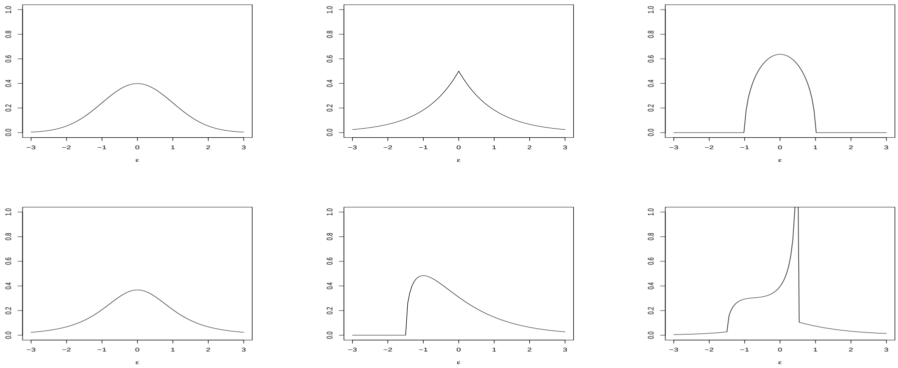
- We discuss more on this later

The Gaussian-noise simple linear regression model[5]

[5] Material in based on the Lecture 4 of Cosma Shalizi's course

The Gaussian-noise simple linear regression model

- If we made more detailed assumptions about the distribution of ε, we could make more precise inference.
- Some possible noise distributions for the simple linear regression model, i.e. $\mathbb{E}[\varepsilon] = 0$



The Gaussian-noise simple linear regression model

- The most common choice is to assume ε follows a Gaussian/Normal distribution.
  - The distribution of X is arbitrary (and perhaps X is even non-random).
  - If $X = x$, then $Y = \beta_0 + \beta_1 x + \varepsilon$, for some constants ("coefficients", "parameters") β_0 and β_1, and some random noise variable ε.
  - $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, independent of X.
  - ε is independent across observations.
- Under this condition then $Y|X = x \sim \mathcal{N}(\beta_0 + \beta_1 x, \sigma^2)$, and Y_i and Y_j are independent given X_i and X_j, so we have

The likelihood

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_i))^2}{2\sigma^2}}$$

and

The Gaussian-noise simple linear regression model

The log-likelihood

$$
\begin{aligned}
\ell(\beta_0, \beta_1, \sigma^2) &= \log(L(\beta_0, \beta_1, \sigma^2)) \\
&= -\frac{n}{2}\log 2\pi - \frac{n}{2}\log \sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_i))^2
\end{aligned}
$$

- Maximum likelihood estimates are derived by taking derivatives and set them to zero.

$$
\begin{aligned}
\frac{\partial \ell}{\partial \beta_0} &= -\frac{1}{2\sigma^2}\sum_{i=1}^{n} 2(y_i - (\beta_0 + \beta_1 x_i))(-1) \\
\frac{\partial \ell}{\partial \beta_1} &= -\frac{1}{2\sigma^2}\sum_{i=1}^{n} 2(y_i - (\beta_0 + \beta_1 x_i))(-x_i)
\end{aligned}
$$

The Gaussian-noise simple linear regression model

$$\begin{aligned}\sum_{i=1}^{n} y_i - (\widehat{\beta_0} + \widehat{\beta_1} x_i) &= 0 \\ \sum_{i=1}^{n} (y_i - (\widehat{\beta_0} + \widehat{\beta_1} x_i)) x_i &= 0\end{aligned}$$

- These are, up to a factor of $1/n$, exactly the equations we got from the method of least squares
- **The least squares solution is the maximum likelihood estimate under the Gaussian noise model**
- The derivative with respect to σ^2:

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2$$

The Gaussian-noise simple linear regression model

- Setting this to 0 we get

$$\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(y_i - (\widehat{\beta_0} + \widehat{\beta_1}x_i))^2 \tag{7}$$

- The solution is the in-sample mean squared error.

An useful result

Suppose that Y_i, $i = 1, \ldots, n$ are n independent random variables with $Y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ and w_i, $i = 0, \ldots, n$, are real constants.
Then
$$w_0 + \sum_{i=1}^{n} w_i Y_i \sim \mathcal{N}\left(w_0 + \sum_{i=1}^{n} w_i \mu_i, \sum_{i=1}^{n} w_i^2 \sigma_i^2\right)$$

Sampling distributions

- Under the Gaussian noise hypothesis we can discuss the sampling distribution of the estimates for $\hat{\beta}_0$ and $\hat{\beta}_1$
- Note that

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \overline{x})(Y_i - \overline{Y})}{\sum_{i=1}^n (x_i - \overline{x})^2} \\
&= \frac{\sum_{i=1}^n (x_i - \overline{x}) Y_i}{\sum_{i=1}^n (x_i - \overline{x})^2} \\
&= \sum_{i=1}^n w_i Y_i
\end{aligned}
$$

Sampling distributions

and

$$
\begin{aligned}
\hat{\beta}_0 &= \overline{Y} - \hat{\beta}_1 \overline{x} \\
&= \frac{1}{n} \sum_{i=1}^{n} Y_i - \overline{x} \sum_{i=1}^{n} w_i Y_i \\
&= \sum_{i=1}^{n} \left(\frac{1}{n} - \overline{x} w_i \right) Y_i \\
&= \sum_{i=1}^{n} w_i^* Y_i
\end{aligned}
$$

- Since both $\hat{\beta}_0$ and $\hat{\beta}_1$ are a linear combination of the Y_i and each Y_i is Gaussianly distributed, then both $\hat{\beta}_0$ and $\hat{\beta}_1$ also follow a Gaussian distribution.

Sampling distributions

- For $\hat{\beta}_0$,

$$\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{x} \sim \mathcal{N}\left(\beta_0,\ \sigma^2 \left(\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^n (x_i - \overline{x})^2}\right)\right).$$

- For $\hat{\beta}_1$ we have,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \overline{x}) Y_i}{\sum_{i=1}^n (x_i - \overline{x})^2} \sim \mathcal{N}\left(\beta_1,\ \frac{\sigma^2}{\sum_{i=1}^n (x_i - \overline{x})^2}\right).$$

Sampling distributions

- Then by standardizing these results using the standard deviation of $\hat{\beta}_i$, $\mathrm{SD}[\hat{\beta}_i]$, we find that

$$\frac{\hat{\beta}_0 - \beta_0}{\mathrm{SD}[\hat{\beta}_0]} \sim \mathcal{N}(0,1) \quad \text{and} \quad \frac{\hat{\beta}_1 - \beta_1}{\mathrm{SD}[\hat{\beta}_1]} \sim \mathcal{N}(0,1)$$

where

$$\mathrm{SD}[\hat{\beta}_0] = \sigma \sqrt{\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}}$$

and

$$\mathrm{SD}[\hat{\beta}_1] = \sigma \frac{1}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}}$$

Sampling distributions

- We don't know σ but we can estimate it using s_e where

$$s_e^2 = \frac{\sum_{i=1}^n e_i^2}{n-2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}$$

- These two new expressions are called

Standard errors

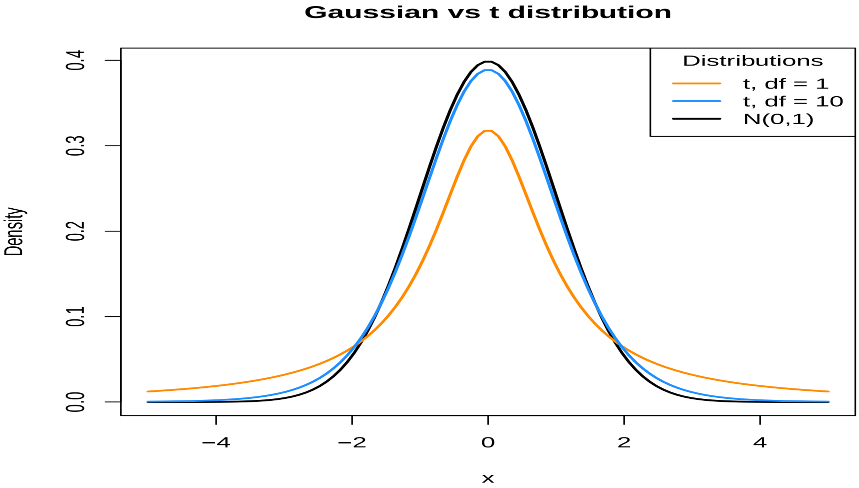
$$\begin{aligned}
\mathrm{SE}[\hat{\beta}_0] &= s_e \sqrt{\frac{1}{n} + \frac{\overline{x}^2}{\sum_{i=1}^n (x_i - \overline{x})^2}} \\
\mathrm{SE}[\hat{\beta}_1] &= s_e \frac{1}{\sqrt{\sum_{i=1}^n (x_i - \overline{x})^2}}
\end{aligned}$$

i.e. the **estimated** standard deviations of the sampling distributions.

Sampling distributions

- If we divide by the standard error, instead of the standard deviation, we obtain the following results which allows us to make confidence intervals and perform hypothesis testing:
$$\frac{\hat{\beta}_0 - \beta_0}{\mathrm{SE}[\hat{\beta}_0]} \sim t_{n-2} \quad \text{and} \quad \frac{\hat{\beta}_1 - \beta_1}{\mathrm{SE}[\hat{\beta}_1]} \sim t_{n-2}$$
where t_d is the t distribution with d degrees of freedom.
- A t distribution is similar to a Gaussian, but with heavier tails.
- As the degrees of freedom increases, the t distribution becomes more and more like a standard Gaussian.
- We can use this result to make inference for the model parameters and the regression function (more on this later)

Sampling distributions



Confidence intervals, prediction intervals and hypothesis tests [6]

[6] Material in these slides is based on chapter 8 of David Dalpiaz's book.

Confidence intervals for β_0 and β_1

- Recall that confidence intervals for means often take the form:

$$\text{est} \pm \text{CRIT} \times \text{SE}$$

  where EST is an estimate for the parameter of interest, CRIT is the critical value, SE is the standard error of the estimate.
- Just some slides ago we have seen the necessary ingredients: SE and the distribution from which we can derive CRIT

Confidence intervals for β_0 and β_1

- Then, for β_0 and β_1 we can create confidence intervals using

$$\hat{\beta}_0 \pm t_{\alpha/2,n-2} \times \mathrm{SE}[\hat{\beta}_0], \quad i.e. \quad \hat{\beta}_0 \pm t_{\alpha/2,n-2} \times s_e \sqrt{\frac{1}{n} + \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

and

$$\hat{\beta}_1 \pm t_{\alpha/2,n-2} \times \mathrm{SE}[\hat{\beta}_1], \quad i.e. \quad \hat{\beta}_1 \pm t_{\alpha/2,n-2} \times \frac{s_e}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

where $t_{\alpha/2,n-2}$ is the critical value such that

$$\Pr(T_{n-2} > t_{\alpha/2,n-2}) = \alpha/2.$$

Hypothesis Testing

- Statistical hypothesis testing: can we falsify the null hypothesis that the parameter of interest has a specific value (or is $\leq$ or $\geq$ than a specific value)
- The null hypothesis typically represent the *status-quo*, the case which brings no change.
- Statistical hypothesis testing assesses whether there is enough evidence to reject the null hypothesis.
- Construction: derive the distribution of an empirical quantity derived from the sample **under the null hypothesis**. If the the empirical quantity is extreme in this distribution it is unlikely that the sample is generated from the hypothetical distribution.

Hypothesis Testing

- Recall that a test statistic (TS) for testing means (like the t-test) often takes the form:

$$\text{TS} = \frac{\text{EST} - \text{HYP}}{\text{SE}}$$

  where EST is an estimate for the parameter of interest, HYP is a hypothesized value of the parameter, and SE is the standard error of the estimate.
- The form above is appropriate when EST follows a gaussian-like distribution with a certain mean (typically the true value of the parameter) and standard deviation SE.
- Then the TS can be compared to a standard gaussian-like distribution. If the test statistic is "large" we reject the null hypothesis: there is enough evidence against the hypothesized value of the parameter.

Hypothesis Testing

- Say we wish to test whether β_0 has a certain value β_0^*.
- The system of hypothesis of interest is

$$H_0 : \beta_0 = \beta_0^* \quad \text{vs} \quad H_1 : \beta_0 \neq \beta_0^*$$

and we use the test statistic

$$t = \frac{\hat{\beta}_0 - \beta_0^*}{\mathrm{SE}[\hat{\beta}_0]} = \frac{\hat{\beta}_0 - \beta_0^*}{s_e \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}}$$

which, under the null hypothesis, follows a t distribution with $n - 2$ degrees of freedom.

Hypothesis Testing

- If we wish instead to test whether the slope β_1 takes a specific value β_1^* the system of hypotheis is

$$H_0 : \beta_1 = \beta_1^* \quad \text{vs} \quad H_1 : \beta_1 \neq \beta_1^*$$

we use the test statistic

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{\text{SE}[\hat{\beta}_1]} = \frac{\hat{\beta}_1 - \beta_1^*}{s_e / \sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}}$$

which again, under the null hypothesis, follows a t distribution with $n-2$ degrees of freedom.

In R

We now return to the penguins example. We use summary to view the results in greater detail. In particular we look at

```
summary(fit)$coefficients
```

```
               Estimate  Std. Error  t value     Pr(>|t|)
(Intercept) 7.03259355 0.457842940 15.36028 1.40630e-35
TV          0.04753664 0.002690607 17.66763 1.46739e-42
```

- Let's now focus on the second row of output, which is relevant to β_1.

```
summary(fit)$coefficients[2,]
```

```
    Estimate   Std. Error      t value     Pr(>|t|)
4.753664e-02 2.690607e-03 1.766763e+01 1.467390e-42
```

- The first value, *Estimate* is $\hat{\beta}_1$

In R

- The second value, *Std. Error*, is the standard error of $\hat{\beta}_1$,

$$\text{SE}[\hat{\beta}_1] = \frac{s_e}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}}$$

- The third value, *t value*, is the value of the test statistic for testing $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$,

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}[\hat{\beta}_1]} = \frac{\hat{\beta}_1 - 0}{s_e / \sum_{i=1}^{n}(x_i - \overline{x})^2}$$

- Lastly, $\Pr(> |t|)$, gives us the p-value of that test.
- The first row of output reports the same values, but for β_0.

Significance of regression, t-Test

- Of course, we could test β_0 and β_1 against any particular value, and perform both one and two-sided tests.
- One very specific test,

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_1 : \beta_1 \neq 0$$

  is used most often and is the default in statistical computing programs.
- For the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i.$$

  if the null hypothesis is true, then $\beta_1 = 0$ and the resulting model reduces to

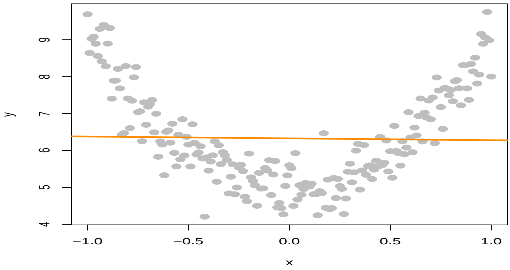
$$Y_i = \beta_0 + \epsilon_i.$$

- In this model, the response does **not** depend on the predictor.
- This implies that there is no relationship between X and Y - no added value in a Simple Linear Regression over taking $\bar{y}$.

Significance of regression, t-Test

- We could think of this test in the following way,
  - Under H_0 there is not a significant linear relationship between X and Y.
  - Under H_1 there is a significance **linear** relationship between x and Y.
- For the `penguins` example, `p-value` ≈ 0
- With this extremely low p-value, we would reject the null hypothesis at any reasonable α level, say for example $\alpha = 0.01$. So we say there is a significant **linear** relationship between mass and flipper length Notice that we emphasize **linear**.

Significance of regression, t-Test

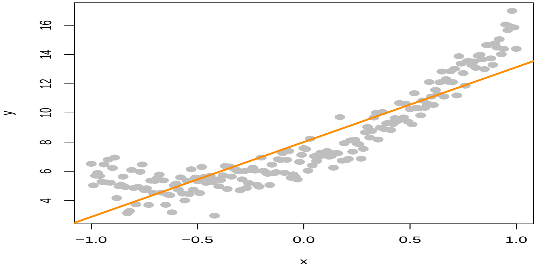


- In this plot of simulated data, we see a clear relationship between x and y, however it is not a linear relationship. If we fit a line to this data, it is very flat.

Significance of regression, t-Test

- The resulting test for $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$ gives a large p-value, in this case 0.75645, so we would fail to reject and say that there is no significant linear relationship between x and y.
- Testing $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$ can only detect straight line relationships.
- On the other hand the significance of the linear terms does not mean that the relationship is entirely linear.

Significance of regression, t-Test



Here the p-value is ≈ 0: we reject H_0, but we shouldn't be entirely satisfied with our model.

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